

Bag of Words (BoW)

Intuition, then advantages and disadvantages.

One Vector per Sentence

1 · ONE HOT ENCODING

A vector for **every word** . Sentence becomes a matrix.



2 · BAG OF WORDS

One vector for the **whole sentence**.

Used for text classification: sentiment analysis, spam or ham.

Dataset, then Lowercase and Remove Stopwords

DATASET		
	Text	O/P
S1	This phone is very fast	1
S2	The camera is fast and sharp	1
S3	Phone camera is fast	1

lowercase +
stopwords



S1 ~~this~~ phone ~~is~~ ~~very~~ fast

S1 → **phone fast**

S2 ~~the~~ camera ~~is~~ fast ~~and~~ sharp

S2 → **camera fast sharp**

S3 phone camera ~~is~~ fast

S3 → **phone camera fast**

"Phone" and "phone" are now the same word.

Vocabulary and Frequency (Highest First)

DATASET · AFTER CLEANING

Text

0/P

S1

phone fast

1

S2

camera fast sharp

1

S3

phone camera fast

1

VOCABULARY · FREQUENCY

fast

3

phone

2

camera

2

sharp

1

WORD	FREQ	FOUND IN
fast	3	S1, S2, S3
phone	2	S1, S3
camera	2	S2, S3
sharp	1	S2

Big datasets: keep only the top N most frequent words as features. Words seen once, like "sharp", can be dropped.

Each Sentence Becomes One Vector

DATASET · AFTER CLEANING

	Text	O/P
S1	phone fast	1
S2	camera fast sharp	1
S3	phone camera fast	1

VOCABULARY · FREQUENCY

fast

3

phone

2

camera

2

sharp

1

	fast	phone	camera	sharp	O/P
S1	1	1	0	0	1
S2	1	0	1	1	1
S3	1	1	1	0	1

1 = word present, 0 = word absent. Columns follow the frequency order.

The Whole Dataset: 3 × 4

DATASET · AFTER CLEANING

	Text	O/P
S1	phone fast	1
S2	camera fast sharp	1
S3	phone camera fast	1

VOCABULARY · FREQUENCY

fast	phone	camera	sharp
3	2	2	1

phone fast S1 = [1, 1, 0, 0] 1 × 4

camera fast sharp S2 = [1, 0, 1, 1] 1 × 4

phone camera fast S3 = [1, 1, 1, 0] 1 × 4

X = [[1, 1, 0, 0], [1, 0, 1, 1], [1, 1, 1, 0]] → shape 3 × 4

TWO VERSIONS

Binary BoW and BoW

SENTENCE "Fast phone, fast camera" → fast phone fast camera

BINARY BOW · 1 OR 0

	fast	phone	camera	sharp
vec	1	1	1	0

Present or not. Count 2 is forced to 1.

BOW · COUNTS

	fast	phone	camera	sharp
vec	2	1	1	0

Count updated by frequency. "fast" = 2.

PART 2

Advantages and Disadvantages of Bag of Words

ADVANTAGES

Simple, and a Fixed Size Input

DATASET · AFTER CLEANING

	Text	O/P
S1	phone fast	1
S2	camera fast sharp	1
S3	phone camera fast	1

VOCABULARY · FREQUENCY

fast	phone	camera	sharp
3	2	2	1

01 Simple and intuitive

02 Fixed size input for ML algorithms

ONE HOT ENCODING
4×7, 4×7, 3×7
size depends on words

BAG OF WORDS
1×4, 1×4, 1×4
size = vocabulary size

DISADVANTAGE 1 OF 4

Sparse Matrix Is Still There → Overfitting

OUR VOCABULARY: 4 WORDS

$S1 = [1, 1, 0, 0]$

REAL VOCABULARY: 50,000 WORDS

$S1 = [0, 0, 1, 0, 0, 0, \dots 0, 1, 0, \dots 0]$ (49,998 zeros)

Mostly zeros, so the model can still overfit.

Ordering of the Words Is Lost

DATASET · AFTER CLEANING

	Text	O/P
S1	phone fast	1
S2	camera fast sharp	1
S3	phone camera fast	1

VOCABULARY · FREQUENCY

fast

3

phone

2

camera

2

sharp

1

S3 was phone camera fast . Its vector follows fast, phone, camera, sharp.

SAME WORDS, OPPOSITE MEANING

"dog bites man" [1, 1, 1]

"man bites dog" [1, 1, 1]

Identical vectors. The model cannot tell them apart.

Out of Vocabulary (OOV) Still Exists

DATASET · AFTER CLEANING

	Text	O/P
S1	phone fast	1
S2	camera fast sharp	1
S3	phone camera fast	1
Test	Phone battery is fast	?

VOCABULARY · FREQUENCY

fast

3

phone

2

camera

2

sharp

1

phone battery fast				
	fast	phone	camera	sharp
Test	1	1	0	0

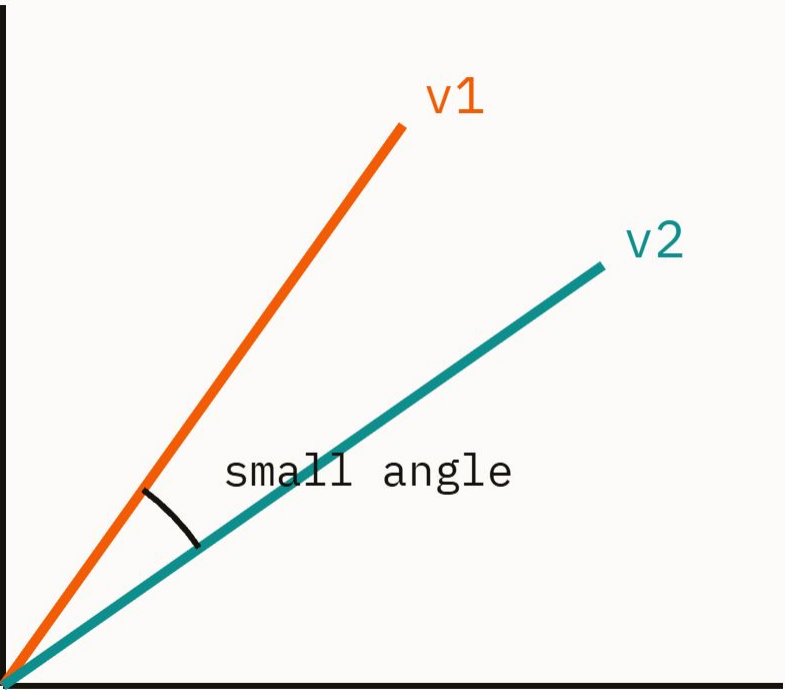
"battery" is not in the vocabulary, so it is **ignored**, even if it matters.

Semantic Meaning Is Still Not Captured

	the	service	was	not	good
v1 · "service was good"	1	1	1	0	1
v2 · "...was not good"	1	1	1	1	1

Cosine similarity $\approx$ **0.89**, so BoW calls them similar. The meanings are opposite.

Every present word gets 1, so the most important word is unknown.



SUMMARY

What to Remember

ADVANTAGES

- 01 Simple and intuitive
- 02 Fixed size input for ML

Binary BoW: 1 or 0. BoW: counts.

DISADVANTAGES

- 01 Sparse matrix → overfitting
- 02 Word order is lost
- 03 Out of vocabulary (OOV)
- 04 No semantic meaning

Next: TF-IDF